Minimizer-Based Mutation Finder

Project report for teammates - status as of 25 Oct 2026. Nothing is implemented in C yet.

1. Summary

- A small tool, written in C with no machine learning, that finds where a virus's genetic text differs from a known correct copy. Each stage is built on a classic data structure from the course (hash table with linked lists, queue made of two stacks, Bloom filter, binary search tree, min-heap, arrays).
- It tells real mutations apart from machine typos by counting how many overlapping reads agree at each position.
- On top of the tool sits a benchmark of how typo rate and settings affect accuracy, speed and memory, to be written up as a conference paper. We do not claim any algorithm is new.

2. The two tracks

Track	Where the sample and answer key come from	What it must handle, and when
Project submission	Our program copies the reference, plants known single-letter changes into the copy, and generates reads from it with random typos. The answer key is exact because we planted the changes.	Single-letter changes only. Partial output due 4 Nov; the full deadline is not known yet.
Research paper (later)	A real Indian SARS-CoV-2 genome is the sample. The answer key is its differences from the reference, found with an existing aligner (Minimap2).	Also missing or extra letters and unknown letters, which the tool must handle. Starts after the project.

Terms used here

Reference. The known correct genome, about 29,900 letters for SARS-CoV-2.

Sample. The patient's genome. We never see it whole.

Read. A short piece (100 to 300 letters in real data) of the sample, in random order.

Mutation. A position where the sample's real letter differs from the reference.

Typo. A wrong letter the machine wrote into a read. It must not be reported as a mutation.

Single-letter change. One letter replaced by another (also called a substitution).

3. Decisions made since the original plan

1. Two tracks, one codebase. The project uses planted single-letter changes. The paper uses a real Indian genome. The same pipeline serves both; the paper needs the extra handling in decisions 2 and 3.

Terms used here

Pipeline. The nine stages from loading the reference to scoring the calls.

2. For the paper, the final counting stage (pile up) will use the alignment path produced by the verify stage. The path says, for each read, which letters match, which are missing and which are extra, so a missing or extra letter no longer shifts every later letter onto the wrong reference position.

Terms used here

Indel. An insertion or deletion: letters are extra or missing compared with the reference. After one, every later letter sits shifted.

Verify stage. Stage 7, which compares all letters of a read with the reference at a candidate start.

Pile up. Stage 8, which counts the letters at each position across all reads and calls a mutation where most reads agree.

Alignment path. The trace through the verify grid. In aligner output it is written as a CIGAR string such as 50M2D30M (50 matched, 2 deleted, 30 matched).

3. Any word that contains an N, even in one letter, is skipped. We believe existing mappers do the same, but this is **not verified yet** and must be checked before the paper states it.

Terms used here

N region. A stretch where the sequencer could not decide the letter and wrote N for unknown. Common in real genomes. N has no 2-bit code.

Word (k-mer). A short fixed-length piece cut from a sequence by sliding one letter at a time. Toy length 4, real about 15.

4. Stage 1 reads the FASTA file one character at a time and writes straight into the 2-bit array, with no intermediate string. The header line and line breaks are ignored.

Terms used here

FASTA. A plain text format for sequences. A header line starts with > and holds the name; the sequence follows, split over lines. The line breaks are not letters. The reference and the Indian genomes come in this format.

2-bit array. A, C, G, T stored as 00, 01, 10, 11, so 16 letters fit in one 32-bit integer.

5. N storage in stage 1. The 2-bit array keeps one cell per position, with a placeholder code (for example 00) under each N. A separate sorted list stores each N run as (start, length) and is checked before any array letter is used, so placeholders are never read as real letters. Putting a marker cell inside the array was rejected: a 2-bit cell cannot hold a marker, and positions would no longer equal indexes.

Terms used here

Placeholder. A valid code written under an N only to keep one cell per position. It has no meaning.

Index equals position. The reference letter at position 7 is read directly from cell 7, with no counting through earlier cells. Later stages do this lookup constantly.

6. Minimap2 is used only in the paper phase: to build the answer key from a real genome, and as a baseline to compare our results against. It is not part of our tool.

Terms used here

Minimap2. An existing, widely used program that finds where one sequence aligns to another.

SAM. The text file format aligners write, one line per aligned sequence. Positions start at 1, while our code counts from 0, so conversions need care.

4. Datasets and tools

What	Link	Status
Reference genome NC_045512.2 (SARS-CoV-2 Wuhan-Hu-1)	https://www.ncbi.nlm.nih.gov/nuccore/NC_045512.2	Proposed. Confirm accession and length on NCBI.
Indian genomes (paper phase): filter Geographic Region to India	NCBI Virus - SARS-CoV-2	No specific genome chosen yet. Accession numbers needed.
Most Indian sequences (INSACOG submissions)	https://gisaid.org	Needs registration and data terms. Optional.
Minimap2	https://github.com/lh3/minimap2	From memory; verify.

5. Open items

- **Implementation.** Decide the rule for when a word overlaps an N run, then write stage 1: the FASTA loader and the N-run list.

- **Literature survey** (due next week). Candidate papers found so far are below. Only one is from IEEE, and it is old (about 2011). No IEEE paper from 2024 or later was found; that search must be run on IEEE Xplore directly. Check each paper's year, venue and relevance before citing it.

Paper	Venue, year	Why it is relevant
Strobealign: flexible seed size enables ultra-fast and accurate read alignment https://link.springer.com/article/10.1186/s13059-022-02831-7	Genome Biology, 2022	Seeding alternative to minimizers, compared against them.
Syncmers are more sensitive than minimizers for selecting conserved k-mers in biological sequences https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7869670/	PeerJ, 2021	Alternative to our word-selection stage.
Theory of local k-mer selection with applications to long-read alignment https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9563685/	Bioinformatics, 2021-22	Trade-offs of picking words, as in stage 2.
New strategies to improve minimap2 alignment accuracy https://arxiv.org/pdf/2108.03515	arXiv, 2021	How a minimizer-based mapper handles very frequent words.
GateSeeder: Near-memory CPU-FPGA acceleration of short and long read mapping https://arxiv.org/pdf/2309.17063	arXiv, 2023	Hash-table index for minimizer seeding.
Recommendations for uniform variant calling of SARS-CoV-2 genome sequence across bioinformatic workflows https://doi.org/10.3390/v16030430	Viruses, 2024	Benchmark framing for virus variant calling.
Investigating sensitivity, specificity and accuracy of variant calling pipelines for analyzing SARS-CoV-2 data https://www.biorxiv.org/content/10.1101/2024.01.24.576385v1.full	bioRxiv preprint, 2024	Compares mapper and caller combinations. Not peer reviewed.
Evaluation of variant calling algorithms for wastewater-based epidemiology using mixed populations of SARS-CoV-2 variants in synthetic and wastewater samples https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10210938/	Microbial Genomics, about 2023	Compares callers on precision and recall with known truth.
Benchmarking Short Sequence Mapping Tools https://ieeexplore.ieee.org/document/6120417/	IEEE, about 2011	Only IEEE result so far; shows how mapper benchmarks are framed.

Suggested IEEE Xplore searches, filtered to 2024 onward: ("variant calling" OR "mutation detection") AND (SARS-CoV-2 OR viral) AND sequencing; minimizer AND ("read mapping" OR seeding); k-mer AND (index OR "Bloom filter") AND genome.